

Car Price Prediction: Detailed Ridge and Lasso Regression Report

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Abstract

This report presents a complete regression workflow for used-car price prediction, including exploratory analysis, feature engineering, and regularised linear modelling. The analysis covers missing-value validation, target transformation, outlier treatment, categorical consolidation, encoding, train-test splitting, scaling, and model diagnostics.

The final modelling stage compares baseline Linear Regression with Ridge and Lasso after leakage correction (excluding raw `price` when predicting `log_price`). Results show stable, moderate predictive performance and consistent behavior across all three models, with regularisation contributing primarily to robustness and validation rather than performance gains.

Keywords: Car Price Prediction, Linear Regression, Ridge, Lasso, Feature Engineering, Multicollinearity

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1 Executive Summary

This report is a detailed write-up of the full notebook workflow used for used-car price prediction. The report is based on the actual notebook outputs and the saved plots generated during the analysis. The work started from raw exploratory analysis, moved through cleaning and feature engineering, and then compared baseline Linear Regression with Ridge and Lasso regularisation.

The corrected outcome of the notebook is that preprocessing decisions mattered more than regularisation, but the true predictive strength of the linear models was moderate rather than near-perfect. The target variable `price` was strongly right-skewed and became much more suitable for modelling after transformation to `log_price`. After removing data leakage by excluding raw `price` from the predictor set, the baseline linear model achieved a realistic test performance around $R^2 \approx 0.785$. Ridge and Lasso confirmed model stability under multicollinearity, but neither materially improved predictive performance.

1.1 Key Results

- Original dataset shape: 15,915 rows and 23 columns.
- Raw target skewness: 1.236.
- Log-transformed target skewness: -0.031.
- Baseline Linear Regression: Train $R^2 = 0.7985$, Test $R^2 = 0.7847$.
- Best Ridge alpha: 0.000800 with best MAE 0.1432.
- Best Lasso alpha: 0.000100 with best MAE 0.1432.
- Ridge Test $R^2 = 0.7847$ and Lasso Test $R^2 = 0.7847$.

2 Problem Statement and Objective

2.1 Objective

The objective of this work was to build a robust regression model for predicting used-car prices and to evaluate whether regularisation improves model performance or interpretability.

2.2 Business Context

Accurate car-price prediction is valuable for resellers, marketplace platforms, and dealerships. In practical terms, such a model can support price benchmarking, feature valuation, and inventory decisions. The notebook therefore focused on both prediction accuracy and interpretability.

2.3 Dataset Overview

- Total rows: 15,915
- Total initial columns: 23
- Primary raw target: `price`
- Modelling target used in regression: `log_price`
- Numerical features identified during EDA: 11
- Categorical features identified during EDA: 9, excluding the multi-feature specification columns

2.4 Modelling Assumptions

The following assumptions were made explicitly in the notebook and are stated here for submission completeness:

- The transformed target `log_price` is more suitable than raw `price` for linear modelling because it substantially reduces skewness.
- When `log_price` is used as the target, raw `price` must be excluded from the predictor set to avoid leakage.
- Outliers were capped rather than removed because extreme prices and usage values may still represent valid market observations.
- Rare categorical levels were consolidated to improve statistical stability and reduce sparse dummy variables.
- Feature scaling was performed only after train-test splitting to avoid leakage.
- High-VIF predictors were retained for Ridge and Lasso so that multicollinearity could be handled through shrinkage rather than by aggressively deleting correlated but informative variables.

3 Notebook Walkthrough: What Was Done and Why

This section follows the notebook flow directly and explains the reasoning behind each step.

3.1 1.1 Data Loading

The dataset was first loaded and inspected using `shape`, `head()`, `info()`, and descriptive statistics. This step validated that the CSV file loaded correctly and gave an immediate sense of the data types and variable ranges.

Finding: the notebook reported the shape as (15915, 23).

3.2 2.1.1 Missing Values Check

The notebook explicitly computed missing-value counts and percentages before any feature engineering or modelling. This was important because hidden missingness can bias downstream transformations and model fitting.

Validation result: the notebook output stated that no missing values were found in the dataset. Because of that, no imputation step was needed.

3.3 2.1.2 Numerical Feature Analysis

Numerical variables were identified and plotted with histograms. Histograms were the correct visual choice here because they show skewness, concentration, long tails, and unusual spread.

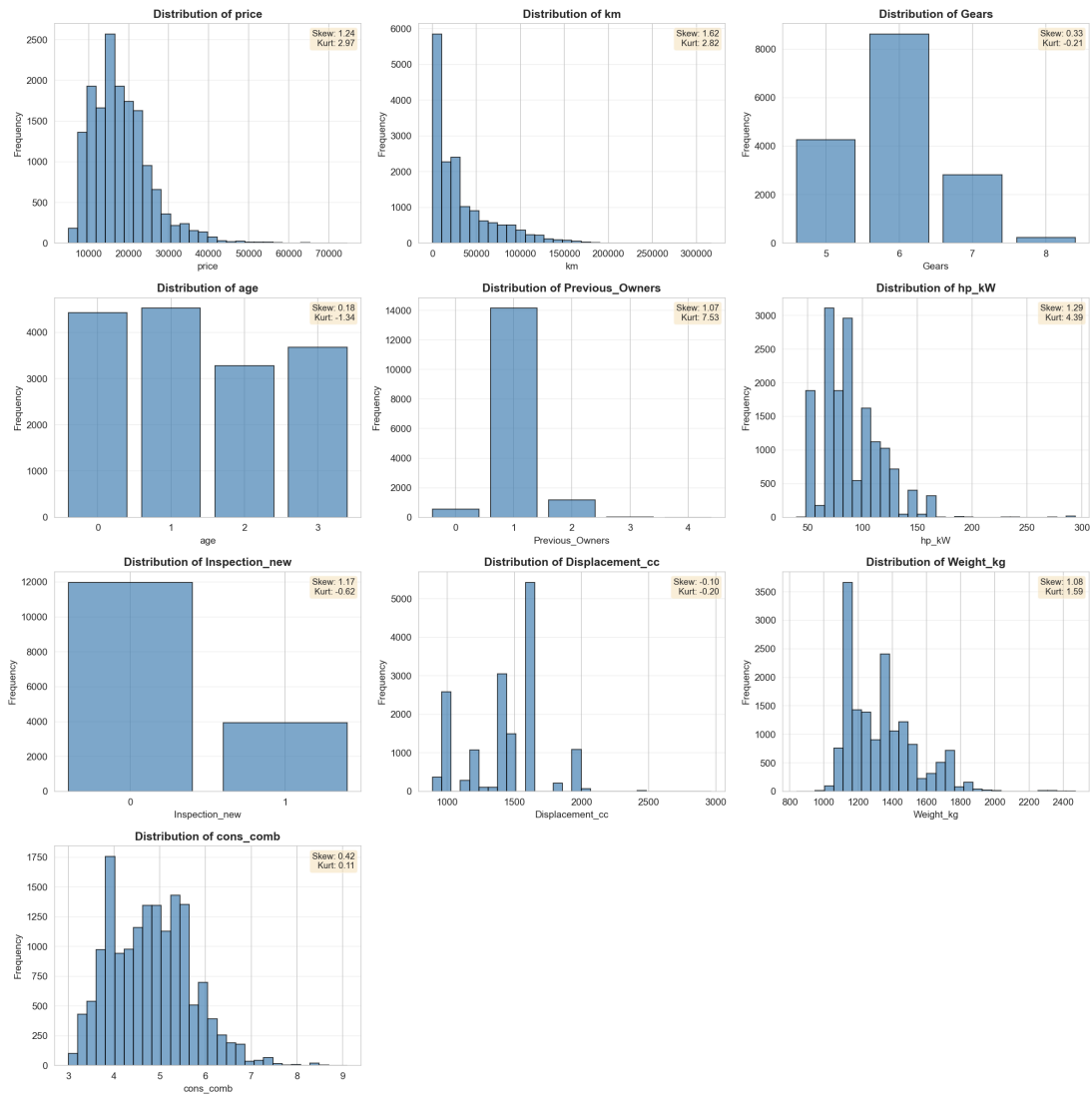


Figure 1: Frequency distributions of the numerical features.

The plots showed that several variables had right-skewed distributions. This was especially important for the target variable and was one of the first signs that transformation would be necessary.

3.4 2.1.3 Categorical Feature Analysis

Categorical variables were counted and visualised with bar charts. This was the correct plot selection because bar charts clearly expose class imbalance and the presence of very small categories.

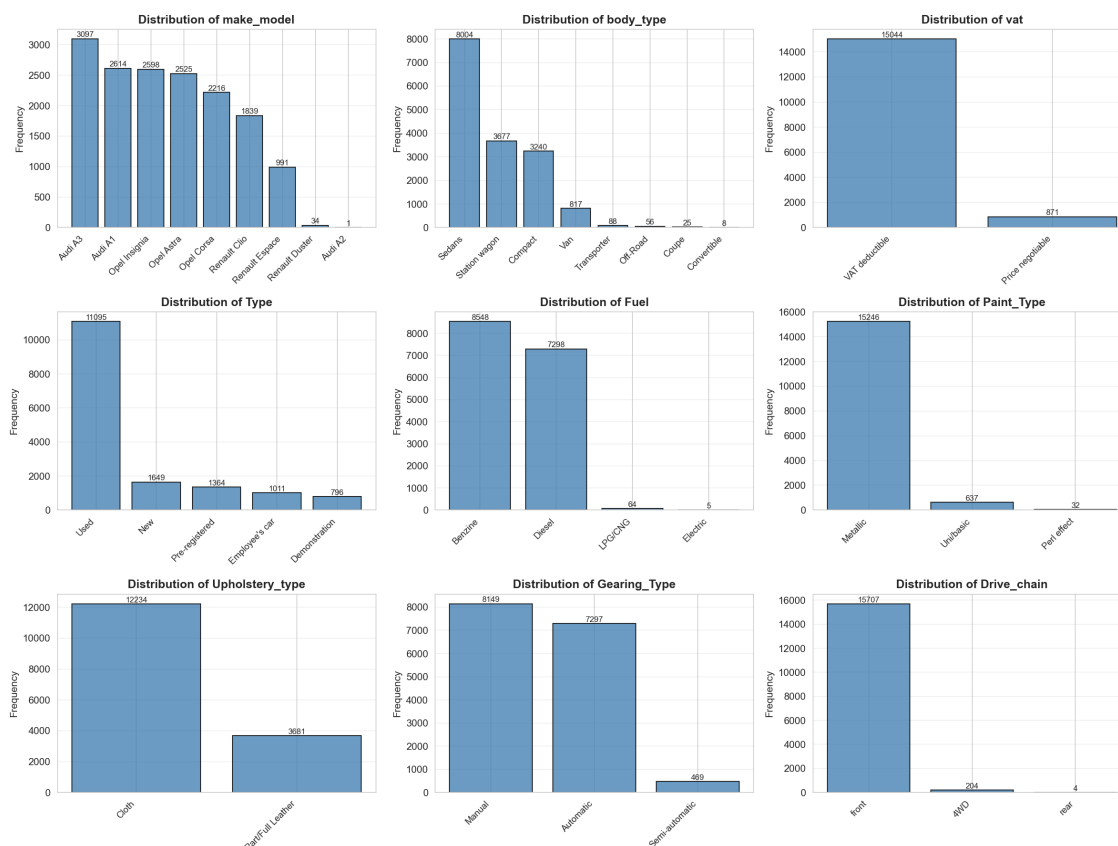


Figure 2: Frequency distributions of the categorical features.

The notebook showed that some variables were strongly imbalanced. For example, `vat` was dominated by `VAT deductible`, and `Drive_chain` was dominated by `front`. These findings later justified consolidation and pruning decisions.

3.5 2.1.4 Category Consolidation and Class Imbalance Handling

The notebook consolidated low-frequency categories and grouped semantically similar vehicle conditions. This was a justified choice because sparse categories increase dimensionality without adding much stable explanatory power.

The most important renaming and regrouping choice was made in the `Type` column. The values `Pre-registered`, `New`, `Employee's car`, and `Demonstration` were grouped into `New/Pre-registered`. This was not an arbitrary name change; it was based on the meaning of those categories as low-use or seller-managed vehicles. The grouping made the feature more interpretable and statistically more stable.

3.6 2.1.5 Target Variable Distribution and Transformation

The notebook used both histogram and boxplot for the target variable. This plot selection was appropriate because the histogram diagnoses skewness while the boxplot highlights

extreme observations.

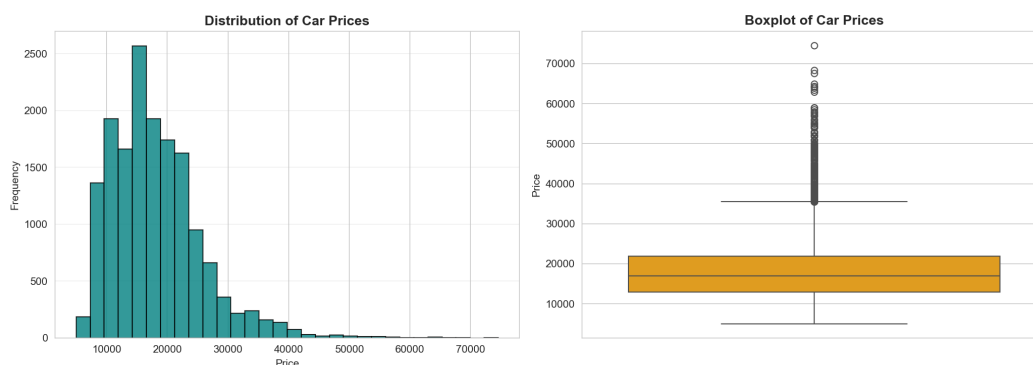


Figure 3: Distribution and boxplot of the raw target variable, `price`.

The notebook reported the following exact values for the raw target:

- Skewness = 1.236
- Kurtosis = 2.966

These numbers, together with the long right tail in the figure, justified applying a log transformation. The notebook created a new target named `log_price` using `log1p(price)`.

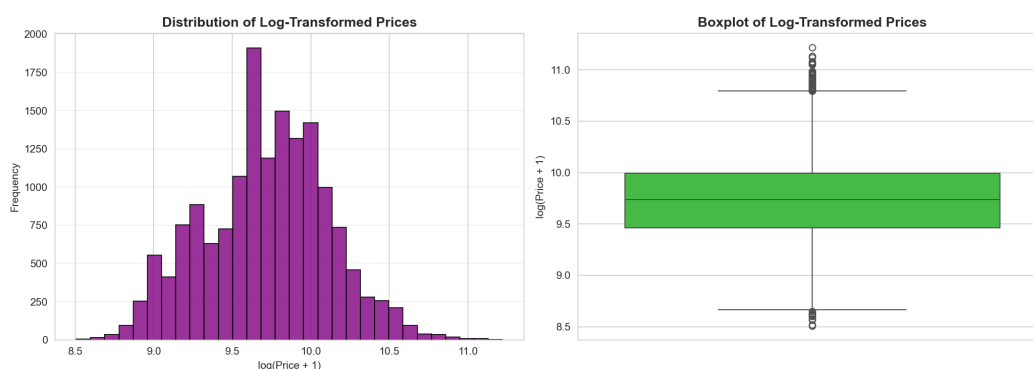


Figure 4: Distribution and boxplot of `log_price` after transformation.

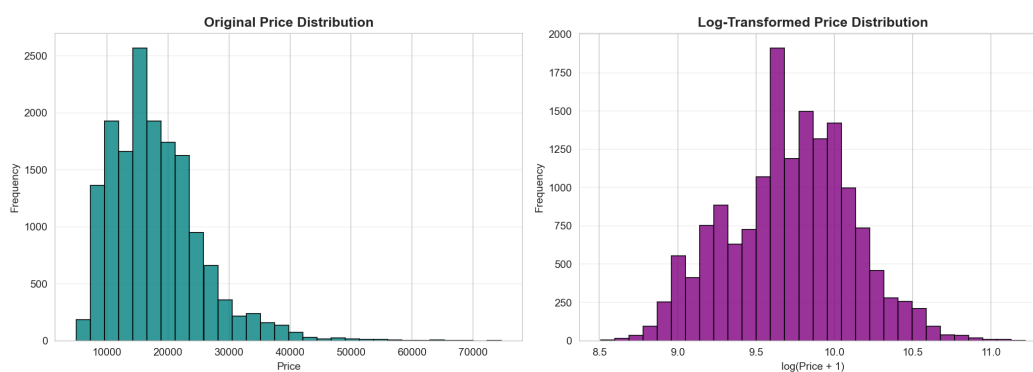


Figure 5: Original versus log-transformed target distribution.

After transformation, the notebook reported:

- Log-transformed skewness = -0.031
- Log-transformed kurtosis = -0.240

This is one of the strongest validations in the entire notebook: the transformation clearly reduced skewness and made the target much more suitable for linear modelling.

4 Correlation and Grouped Comparisons

4.1 2.2.1 Correlation Analysis

The notebook used a correlation heatmap to inspect relationships among numerical variables and the transformed target.

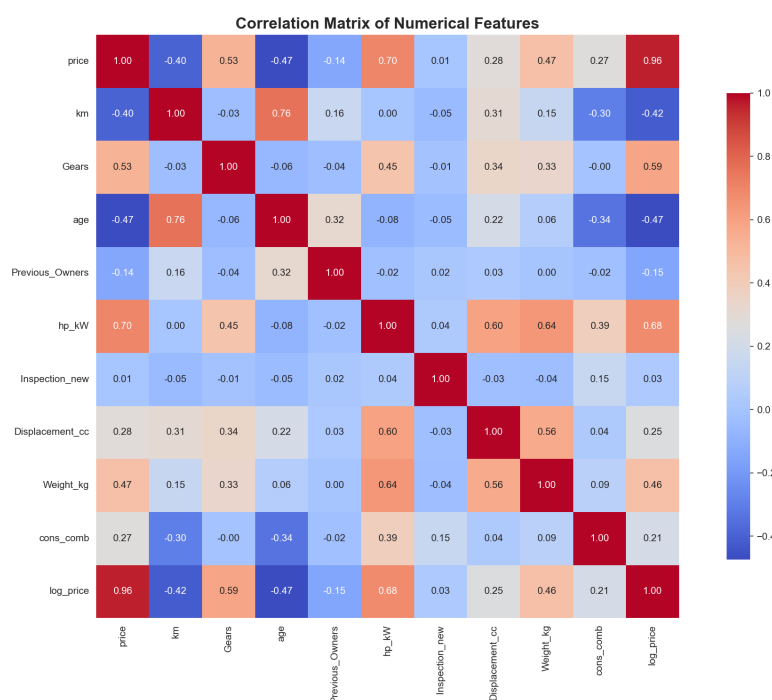


Figure 6: Correlation matrix including log_price.

The printed correlation results showed the following strongest relationships with log_price:

- hp_kW: 0.678024
- Gears: 0.588194
- age: -0.474618
- Weight_kg: 0.464597

- km: -0.419189
- Displacement_cc: 0.254854

These findings are aligned with business intuition. More powerful and heavier cars tend to be more expensive, while older cars and cars with more mileage tend to be cheaper.

4.2 2.2.2 Categorical Feature Versus Target Analysis

The notebook also plotted mean `log_price` by categorical groups. These bar plots were a strong choice because they directly answer whether some categories tend to sell at a higher price level than others.

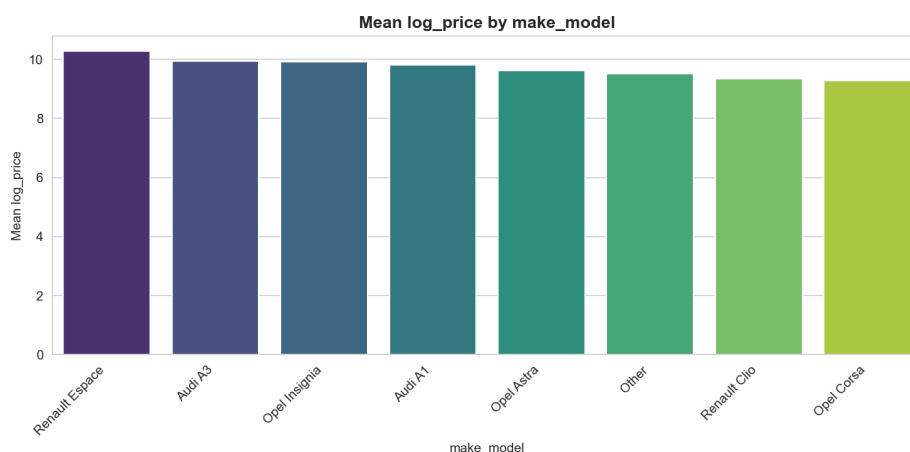


Figure 7: Mean transformed price by model grouping.

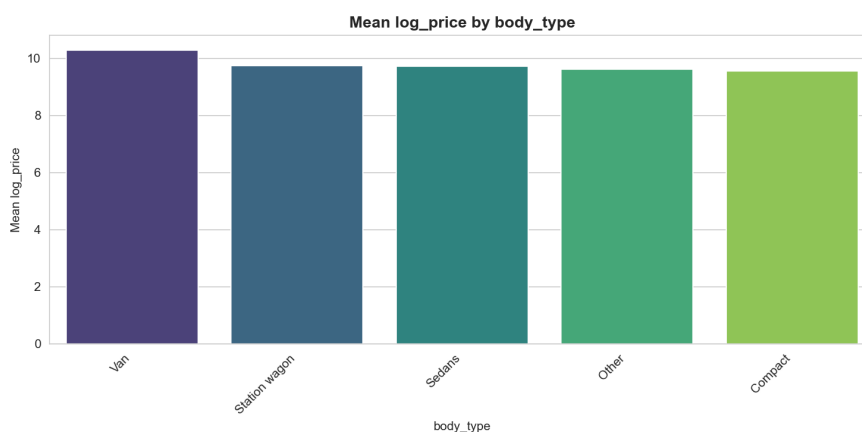


Figure 8: Mean transformed price by body type.

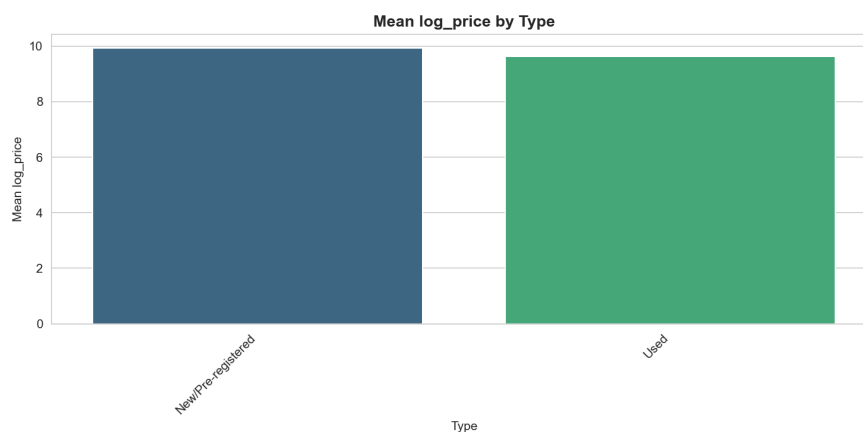


Figure 9: Mean transformed price by vehicle condition type.

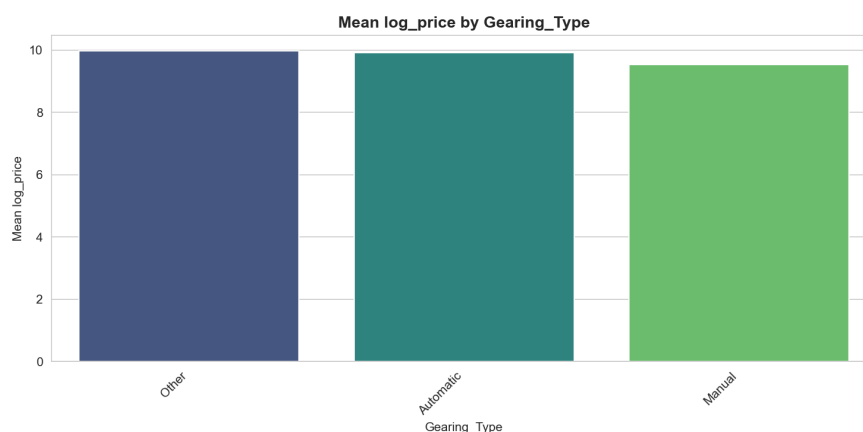


Figure 10: Mean transformed price by gearing type.

Findings from these grouped plots:

- **Renault Espace** appeared as the highest-priced grouped model category on average.
- **Van** had the highest mean transformed price among body types.
- **New/Pre-registered** vehicles were priced above **Used** vehicles on average.
- **Automatic** and **Other** gearing categories were priced above **Manual**.
- Categories such as **Part/Full Leather** and non-front **Drive_chain** also showed price premiums in their dedicated plots.

These outputs justified retaining the major categorical signals in the analytical story and showed that price differences were not random across groups.

5 Outlier Analysis and Handling

5.1 2.3.1 Outlier Detection

The notebook used boxplots together with IQR-based outlier counts. This was a sound validation method because boxplots provide visual evidence while the IQR rule provides a reproducible numerical rule.

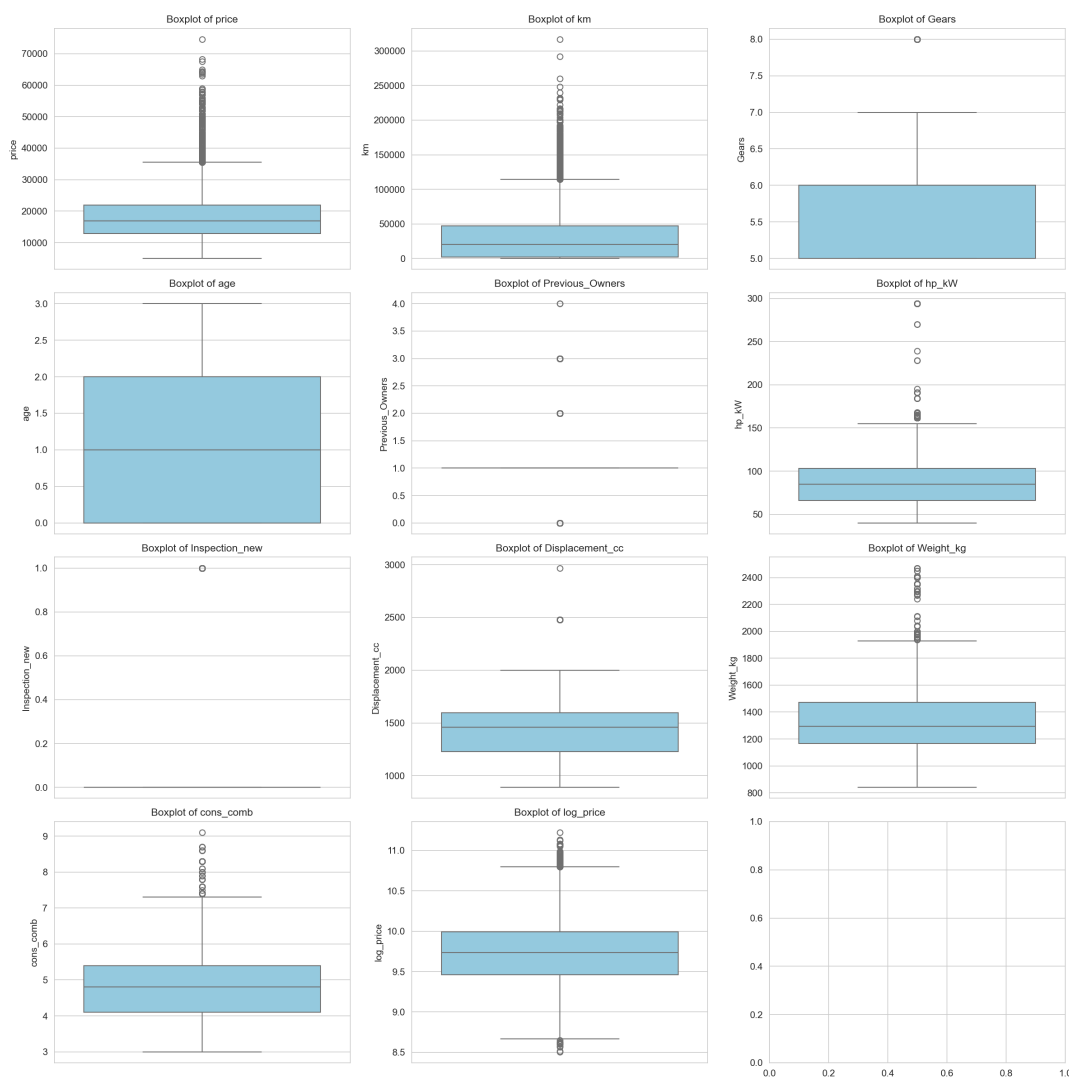


Figure 11: Numerical feature boxplots before capping.

The most notable outlier counts reported by the notebook were:

- price: 479 outliers (3.01%)
- km: 689 outliers (4.33%)
- Previous_Owners: 1,757 outliers (11.04%)

- `Inspection_new`: 3,932 outliers (24.71%)
- `log_price`: 71 outliers (0.45%)

5.2 2.3.2 Outlier Treatment

The notebook capped outliers using the IQR method rather than deleting entire rows. This was an appropriate choice because it reduces the influence of extreme values while preserving the overall sample size and market coverage.

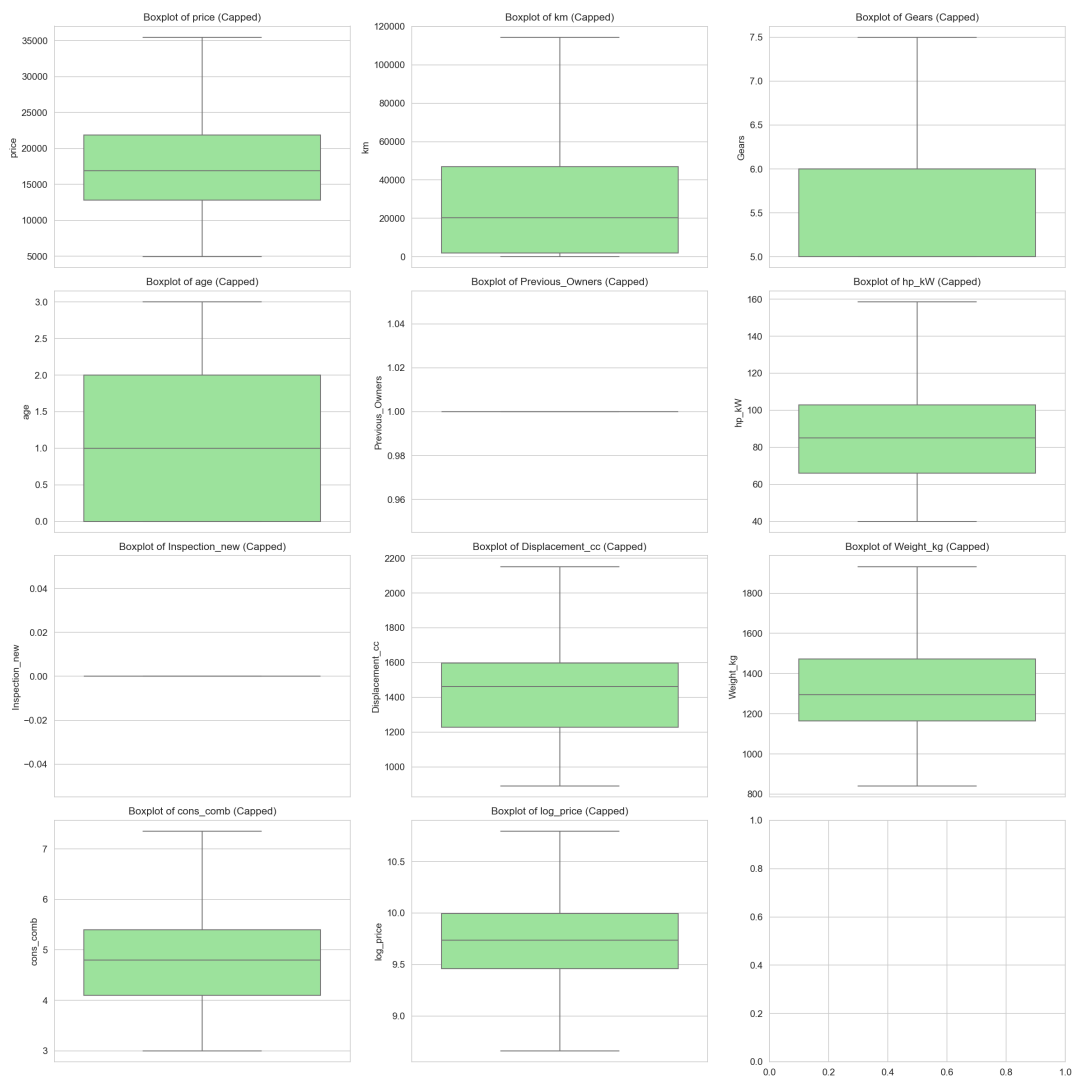


Figure 12: Numerical feature boxplots after capping.

The notebook reported the following capped counts for the same variables:

- `price`: 479 values capped
- `km`: 689 values capped
- `Previous_Owners`: 1,757 values capped

- `Inspection_new`: 3,932 values capped
- `log_price`: 71 values capped

The capping choice was justified because the task was price prediction, not anomaly removal, and retaining observations is usually preferable in a large commercial dataset.

6 Feature Engineering, Discarding, and Encoding

6.1 2.4.1 Redundant Columns and Name Changes

The notebook split `make_model` into two additional variables, `make` and `model`. This name change was justified because it separates brand-level information from model-level information and makes the categorical structure more interpretable.

The notebook also checked for structural cleanup issues and reported:

- No constant columns to drop.
- No duplicate rows found.

6.2 2.4.2 Specification Columns

The columns `Comfort_Convenience`, `Entertainment_Media`, `Extras`, and `Safety_Security` contain comma-separated lists rather than single categories. The notebook examined these columns separately because they cannot be treated as standard categorical variables without losing meaning.

The feature-engineering logic attempted to convert these lists into machine-usable indicators and to remove indicators that were too rare or too common. This was a justified design choice because such feature-list columns are inherently multi-label.

Discarding logic used in the notebook:

- Candidate specification features were filtered if they appeared in fewer than 5% or more than 95% of vehicles.
- After encoding, an additional pruning pass removed binary indicators with prevalence below 2% or above 98%.

The notebook then reported a final DataFrame shape after specification-feature pruning of (14198, 24).

6.3 2.4.3 Encoding

The remaining categorical variables were one-hot encoded. This was the correct general approach for linear models because one-hot encoding preserves category identity without creating artificial numeric order.

The notebook reported that after encoding, the DataFrame shape became (14198, 11671).

6.4 2.4.4 Train-Test Split

The dataset was split into training and testing partitions using an 80:20 split and `random_state = 42`.

Observed result:

- Training set: 11,358 rows
- Test set: 2,840 rows

This split was justified because it preserves a large training sample while keeping a sufficiently large holdout set for evaluation.

6.5 2.4.5 Scaling

Scaling was done with `StandardScaler` after the train-test split. This sequencing prevented data leakage and was especially important because both Ridge and Lasso penalise coefficients based on scale.

The notebook reported scaled array shapes of (11358, 7) for training and (2840, 7) for testing after removing raw `price` from the predictor set.

7 Baseline Linear Regression Model

7.1 3.1.1 Baseline Model Fit

The baseline model was fitted first to establish a reference point for all regularised models. This was an important modelling choice because regularisation should only be claimed as useful if it improves or stabilises a strong baseline.

The exact baseline metrics reported by the corrected notebook were:

Table 1: Baseline linear regression metrics on the transformed target.

Metric	Train	Test
R^2	0.7985	0.7847
RMSE	0.18	0.18
MAE	0.14	0.14
MSE	0.03	0.03

*Note: these errors are on the **log_price** scale because the transformed target was used during modelling.*

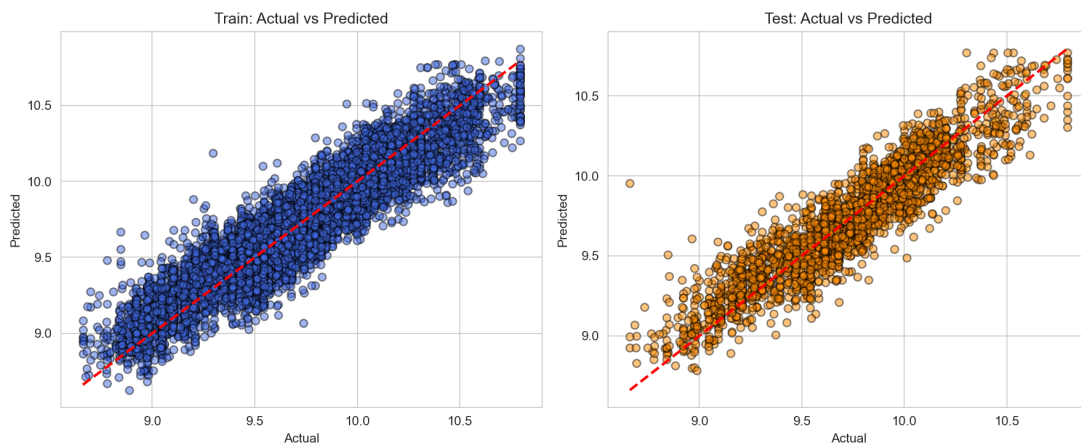


Figure 13: Actual versus predicted values for the baseline linear model.

The actual-versus-predicted figure still showed a clear linear relationship, but with visibly more spread around the reference line than in the leakage-affected version. This is consistent with a realistic R^2 around 0.785 rather than an artificially near-perfect fit.

7.2 3.1.2 Residual Validation

The notebook validated the regression assumptions using residual plots. This was a correct step because high predictive accuracy alone does not confirm that a linear model is well-behaved.

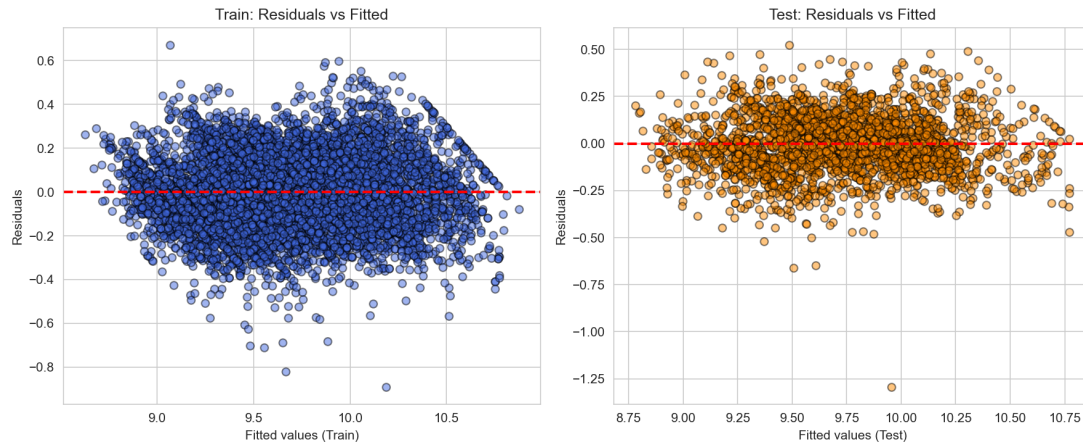


Figure 14: Residuals versus fitted values for train and test sets.

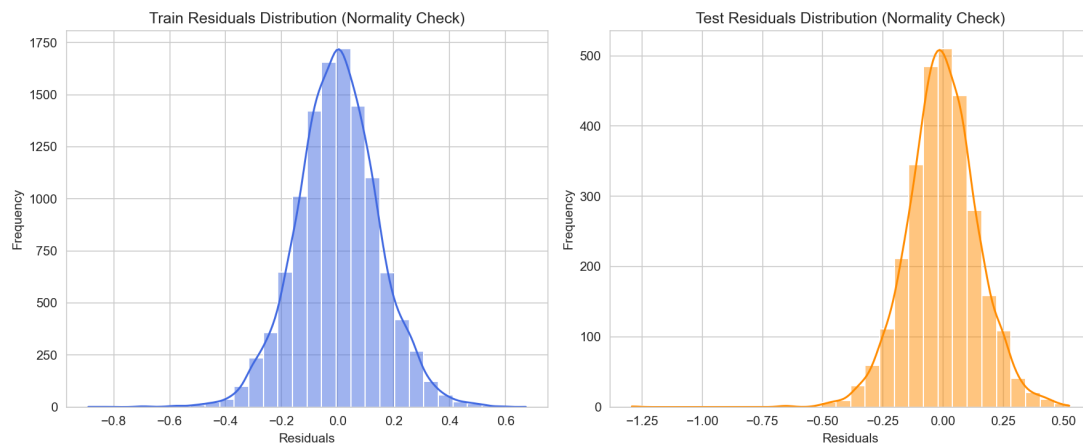


Figure 15: Residual histograms for train and test sets.

The residual plots suggested the following:

- **Linearity:** only approximate, because the residuals versus fitted plot showed visible curvature rather than a purely random cloud.
- **Normality:** reasonably acceptable after the target transformation, with residuals centered near zero.
- **Homoscedasticity:** not perfect, because the residual spread changed across the fitted range.
- **Overfitting:** limited but present, because the train and test R^2 values differed by about 0.0138.

7.3 Multicollinearity Validation

The notebook calculated VIF values on the corrected numeric predictor set and identified a set of features with VIF above 10. The following variables were flagged in the notebook output:

- Gears
- hp_kW
- Displacement_cc
- Weight_kg
- cons_comb

The notebook retained these variables for the regularised models instead of dropping them manually. This was the chosen handling strategy because the assignment explicitly involved Ridge and Lasso, which are designed to stabilise models under multicollinearity.

8 Ridge Regression

8.1 3.2.1 Coarse Alpha Search

The notebook searched Ridge alpha over a wide log-spaced range from 0.001 to 1000. Using a wide log grid was a justified choice because regularisation effects can change slowly at first and then sharply when the penalty becomes very large.

The notebook reported:

- Best coarse alpha = 0.0010
- Best coarse Test $R^2 = 0.7847$
- Best coarse MAE = 0.1432

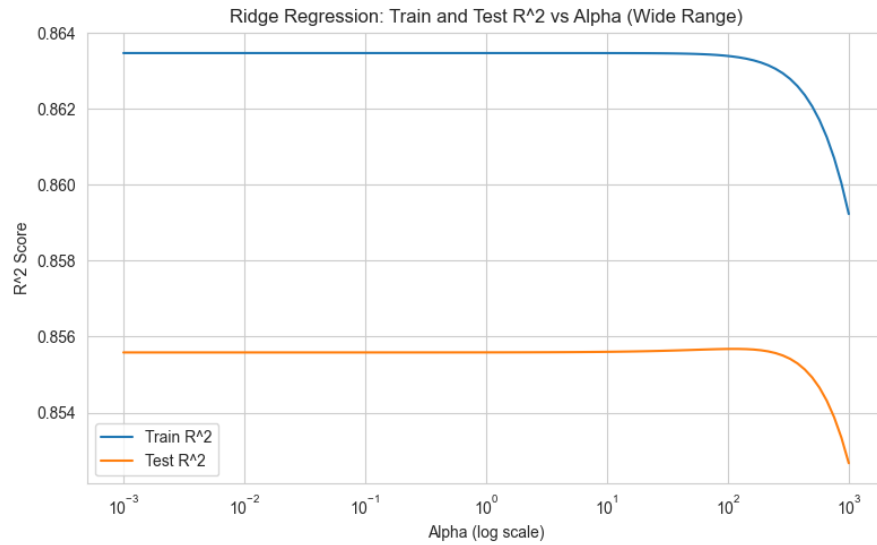


Figure 16: Ridge train and test R^2 across the wide alpha grid.

This figure showed that performance was almost unchanged over the useful part of the alpha range and deteriorated only when alpha became very large. That indicates the model was already fairly stable and not highly sensitive to the Ridge penalty near the optimum.

8.2 3.2.2 Fine Alpha Search

The notebook then refined the search around the coarse optimum. This was the correct follow-up because a narrow local search gives a more precise alpha estimate once the correct region has been identified.

The notebook reported:

- Best fine-tuned alpha = 0.000800
- Best fine-tuned MAE = 0.1432
- Ridge Test R^2 = 0.7847



Figure 17: Ridge train and test R^2 in the fine alpha range.

The fine-range figure was essentially flat, which means regularisation strength had almost no practical influence near the optimum.

8.3 Ridge Coefficient Interpretation

The final Ridge model was fitted with the selected alpha and its coefficients were inspected in the notebook. The main interpretation remained unchanged: mileage retained a negative effect, and age remained an important pricing signal after shrinkage. This is consistent with the earlier correlation findings.

9 Lasso Regression

9.1 3.3.1 Coarse Alpha Search

The notebook used a wide Lasso grid from 10^{-4} to 10^2 . This was a justified search design because Lasso can move from a useful sparse model to a collapsed model very quickly when alpha increases.

The notebook reported:

- Best coarse alpha = 0.000100
- Best coarse MAE = 0.1432

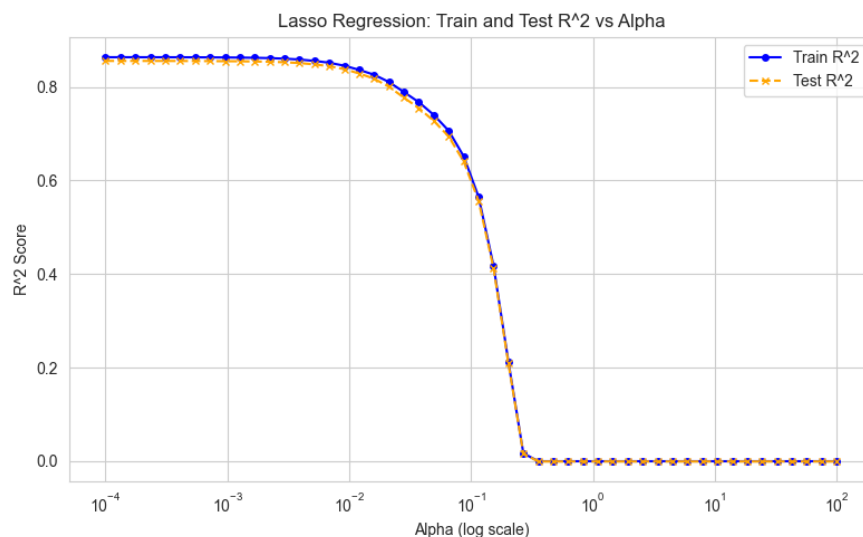


Figure 18: Lasso train and test R^2 across the wide alpha range.

The wide-range Lasso plot was particularly informative. It showed that performance stayed strong at very low alpha values but collapsed sharply at larger alpha values. This is typical Lasso behaviour when the penalty becomes strong enough to eliminate too much information.

9.2 3.3.2 Fine Alpha Search

The notebook then refined the search in a narrow range around the best coarse alpha.

The notebook reported:

- Best fine-tuned alpha = 0.000100
- Best fine-tuned MAE = 0.1432
- Lasso Test $R^2 = 0.7847$

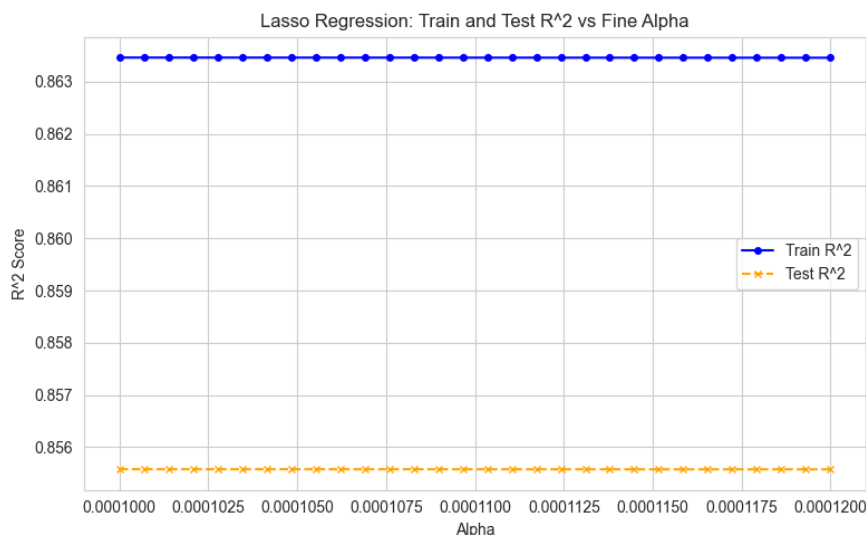


Figure 19: Lasso train and test R^2 in the fine alpha range.

9.3 Lasso Coefficient Interpretation and Feature Elimination

The final Lasso model was fitted and its coefficients were printed. The notebook also listed features with zero coefficients. This confirmed that Lasso was performing feature selection, but the key finding was that the sparsity benefit did not improve prediction quality compared with Ridge or the baseline model.

10 Model Comparison and Final Validation

10.1 Side-by-Side Performance Summary

Table 2: Summary of model performance from notebook outputs.

Model	Train R^2	Test R^2	Best Alpha	Best MAE
Linear Regression	0.7985	0.7847	Not applicable	0.14 (test MAE)
Ridge Regression	≈ 0.7985	0.7847	0.000800	0.1432
Lasso Regression	≈ 0.7985	0.7847	0.000100	0.1432

Main comparison finding: all models were effectively tied on predictive performance after leakage was removed. This means regularisation did not improve the score; it mainly confirmed that the fitted solution was already stable under multicollinearity.

10.2 Validation of Choices Made in the Notebook

Table 3: Validation summary for the notebook choices.

Choice	Validation Status	Evidence and Justification
Log transformation of target	Strongly validated	Reduced skewness from 1.236 to -0.031 and visibly improved the target distribution.
Outlier capping	Validated	Reduced the influence of long-tail values while preserving sample size.
Category consolidation	Validated	Made sparse categories more stable and more interpretable.
Scaling after split	Validated	Prevented leakage and ensured fair Ridge/Lasso penalties. Raw <code>price</code> was also excluded from predictors when modelling <code>log_price</code> .
Testing Ridge	Validated	Multicollinearity was present, so shrinkage-based validation was justified.
Testing Lasso	Validated	Useful for sparsity checks, though not superior on test performance.
Linear model sufficiency	Partially validated	Performance remained reasonable after leakage removal, though residual curvature and only moderate R^2 indicate that a linear model is adequate but not exhaustive.

11 Recommendations

11.1 Recommended Model

The best practical recommendation from the notebook is **Ridge Regression with alpha 0.000800**. The reason is not that Ridge clearly outperformed the other models, but that it matched the best observed performance while being the safest choice when multicollinearity is present.

If absolute simplicity is preferred, the baseline linear model is also acceptable because its metrics were essentially identical.

11.2 Business Recommendations

The notebook suggests the following practical interpretations:

- Vehicle power and transmission-related characteristics contribute positively to price.
- Age and mileage remain strong negative pricing signals.
- Vehicle segment and model grouping matter, as shown in the grouped mean-price plots.

- Premium configurations such as better upholstery and automatic gearing tend to be associated with higher average prices.

11.3 Limitations and Future Work

This notebook also highlighted some limitations that should be stated in the report:

- The residual versus fitted plots showed curvature, so the linearity assumption is not perfect.
- Error metrics were computed on the transformed target scale.
- The specification-list columns are structurally complex and may benefit from cleaner sparse encoding.
- External market factors and time effects were not included.

Future work could include Elastic Net, tree-based models, and richer market covariates.

12 Final Conclusion

The strongest finding from the notebook was not that regularisation dramatically improved performance, but that careful preprocessing was necessary to obtain a realistic and stable linear solution. The target transformation was the single most important modelling decision because it corrected the right-skewed price distribution and made the target much more compatible with linear regression. The outlier analysis, category consolidation, feature engineering, and scaling steps further stabilised the pipeline.

The exploratory analysis showed a clear economic story: higher power, higher weight, and higher-spec vehicle groups tend to be priced higher, while age and mileage reduce value. The grouped plots supported those conclusions across model families, body types, gearing types, and vehicle conditions.

Once preprocessing was complete and raw `price` was excluded from the predictor set, the baseline linear model achieved train and test R^2 values of 0.7985 and 0.7847 respectively. Ridge and Lasso confirmed that there was only limited overfitting and little sensitivity to regularisation in the useful alpha range. Ridge remained the most defensible final recommendation because it provides coefficient stabilisation without sacrificing accuracy, but the notebook as a whole shows that linear models explain a substantial part of price variation rather than nearly all of it.

Appendix: Additional Generated Figures

The submission instructions asked that the graphs generated during analysis should be attached to the report. The following saved figures were also generated in the notebook and are included here for completeness.

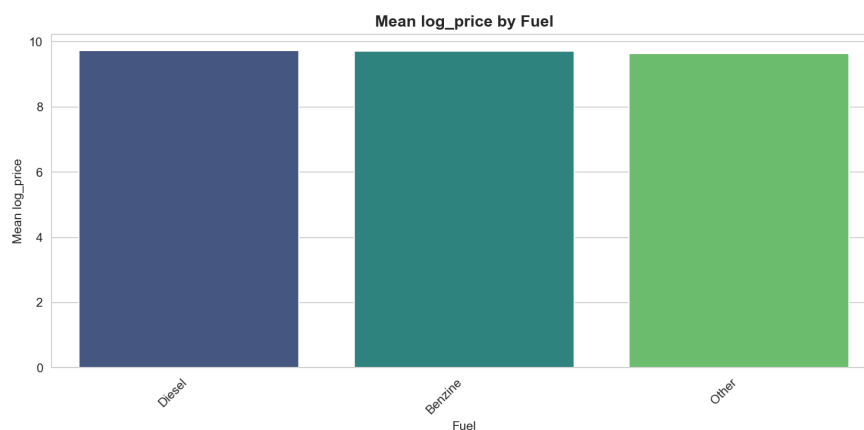


Figure 20: Mean transformed price by fuel type.

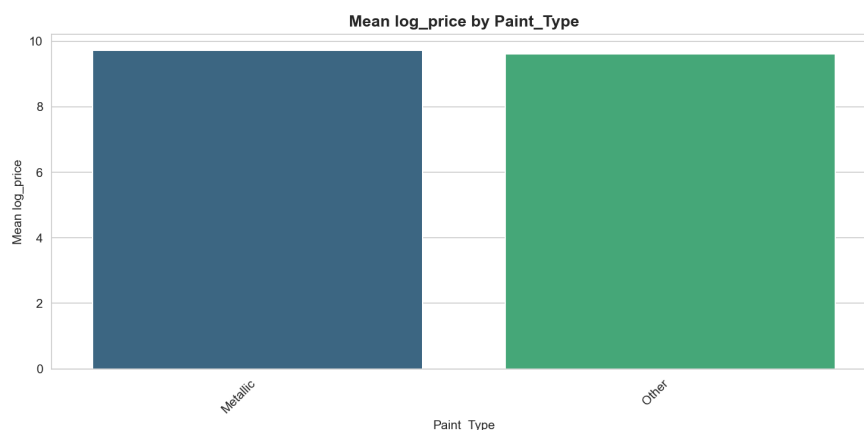


Figure 21: Mean transformed price by paint type.

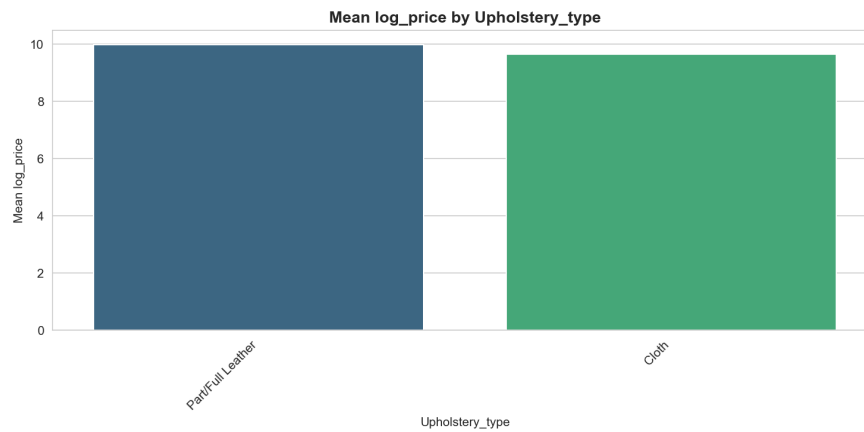


Figure 22: Mean transformed price by upholstery type.

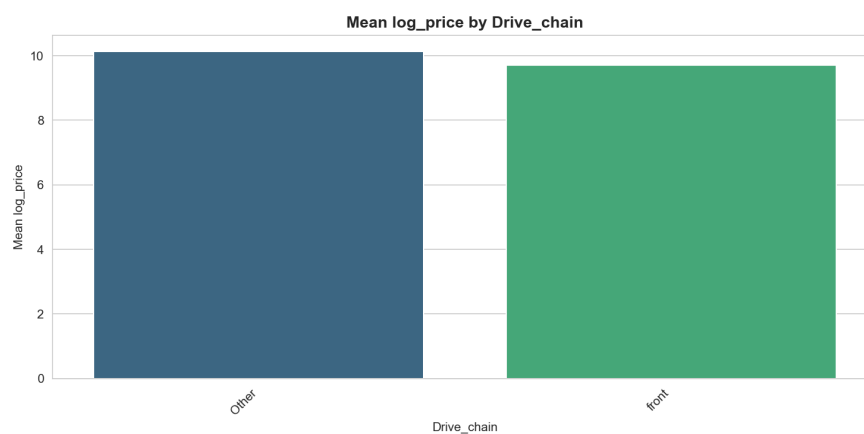


Figure 23: Mean transformed price by drive chain.



Figure 24: Mean transformed price by VAT category.

References

1. Hastie, T., Tibshirani, R., and Friedman, J. *The Elements of Statistical Learning*. Springer.
2. James, G., Witten, D., Hastie, T., and Tibshirani, R. *An Introduction to Statistical Learning*. Springer.
3. Scikit-learn documentation for Linear Regression, Ridge, and Lasso: https://scikit-learn.org/stable/modules/linear_model.html